Experiment 9

Aim

Study and implement Identity and Access Management (IAM)

Theory

1)Study Identity and Access Management (IAM)

Identity and Access Management (IAM) is a security framework used in cloud computing to manage digital identities and control access to resources. It ensures that only authorized users can access specific services and perform permitted operations. In a cloud environment where multiple users interact with shared resources, IAM plays an important role in maintaining security and protecting sensitive information.

Identity management focuses on identifying and authenticating users in the system. This process verifies the identity of users through credentials such as usernames, passwords, or security tokens. Once the identity is verified, access management determines what resources the user is allowed to access and what actions they are permitted to perform. This combination of authentication and authorization helps prevent unauthorized access.

Cloud platforms implement IAM using different components such as users, groups, roles, and policies. A user represents an individual who needs access to the system. Groups allow administrators to manage permissions for multiple users at once. Roles provide temporary permissions that can be assigned to users or services. Policies define the rules that specify which actions are allowed or denied.

For example, AWS Identity and Access Management allows administrators to create users and assign permission policies to control access to AWS resources. By implementing IAM correctly, organizations can ensure that users only have access to the resources required for their work, which improves security and reduces the risk of misuse.

2)Need for Access Management Services in Cloud Computing

Access management services are essential in cloud computing because cloud environments often contain large amounts of sensitive data and critical applications. Without proper access control mechanisms, unauthorized users could gain access to confidential information or misuse system resources.

One of the main reasons access management is required is to protect sensitive data from unauthorized access. Organizations store important information such as customer data, financial records, and business applications in the cloud. Access management ensures that only authorized individuals can view or modify this information.

Another important reason is authentication and authorization. Authentication verifies the identity of the user attempting to access the system, while authorization determines what resources the user is allowed to access. These processes help maintain system integrity and prevent unauthorized activities.

Access management also helps organizations comply with security regulations and policies. Many industries require strict control over user access to protect sensitive data. IAM systems provide monitoring and logging capabilities that allow administrators to track user activities and ensure compliance with security standards.

Additionally, access management services improve administrative efficiency by allowing centralized control of permissions. Instead of managing permissions for each individual user separately, administrators can assign roles and policies to groups of users. Cloud providers such as Amazon Web Services offer advanced IAM tools that simplify this process and enhance security.

Activity

Case Study Demonstration of Identity and Access Management (IAM) on AWS

Consider a company that uses cloud services to host its applications and store its data. The company has multiple employees such as developers, database administrators, and system administrators. Each employee needs access to specific cloud resources depending on their responsibilities. To manage this securely, the company uses IAM.

First, the administrator creates individual user accounts in AWS Identity and Access Management for each employee. These accounts allow users to log in to the AWS management console using their credentials.

Next, the administrator creates groups such as Developers, Database Administrators, and System Administrators. Each group is assigned specific permissions based on job roles. For example, developers may have permission to deploy applications on Amazon EC2, while database administrators manage databases using Amazon RDS.

Policies are then attached to these groups to define what actions the users are allowed to perform. For instance, developers may have permission to start or stop servers but not delete critical infrastructure. This ensures that users only have access to the resources required for their tasks.

Another important concept used in IAM is roles. Roles allow temporary access to services without storing permanent credentials. For example, an EC2 instance may need access to files stored in Amazon S3. Instead of storing access keys in the application, an IAM role is assigned to the EC2 instance, allowing secure access to the storage service.

AWS also supports Multi-Factor Authentication (MFA), which requires users to provide an additional verification code during login. This adds another layer of security and reduces the risk of unauthorized access.

Through this case study, it becomes clear that IAM helps organizations manage user identities, control permissions, and secure cloud resources effectively.

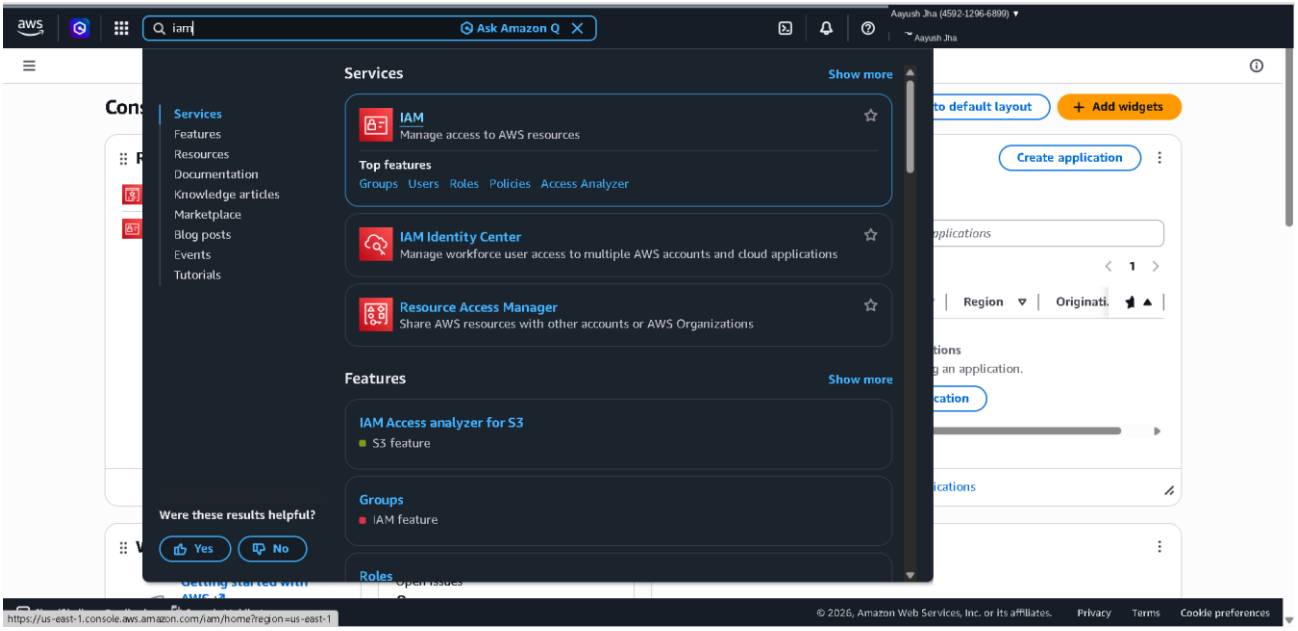
Conclusion

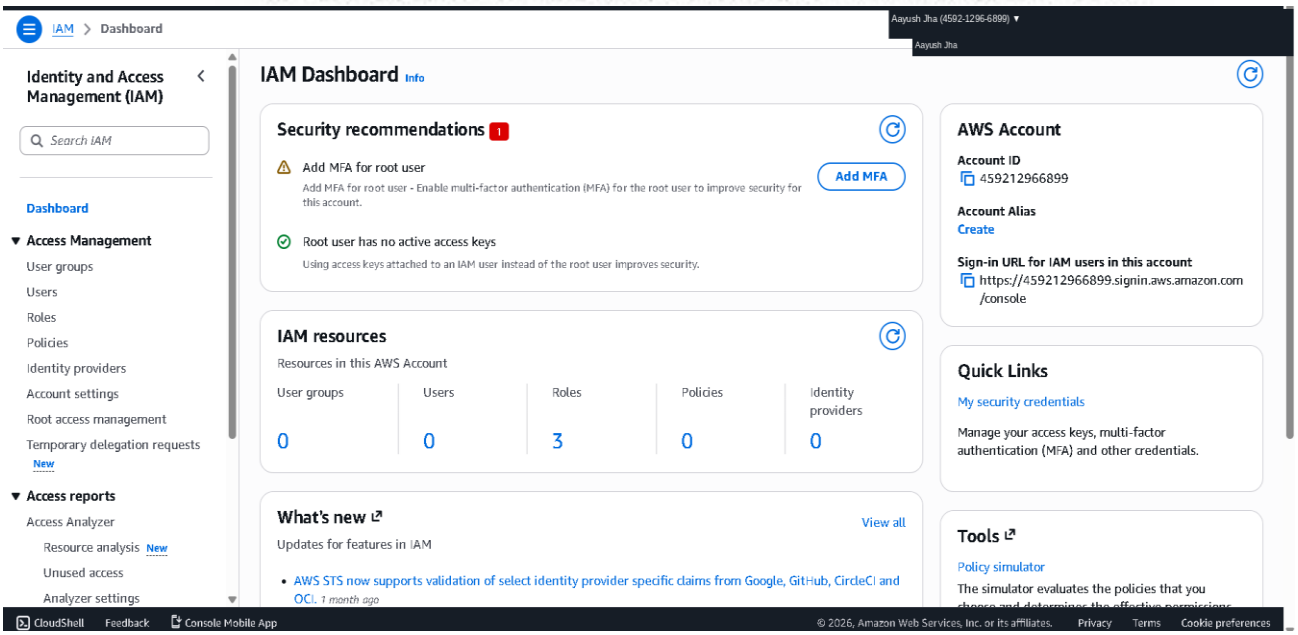
Identity and Access Management is a critical component of cloud security because it controls how users interact with cloud resources. By managing identities and permissions properly, organizations can protect their systems from unauthorized access and security threats.

Cloud platforms such as Amazon Web Services provide powerful IAM services that allow administrators to manage users, groups, roles, and policies efficiently. These tools help organizations implement strong security practices and ensure safe operation of cloud infrastructure. Therefore, IAM plays an essential role in maintaining the security and reliability of cloud computing environments.

References

1. Amazon Web Services Official Documentation
2. National Institute of Standards and Technology Cloud Security Guidelines
3. Cloud Security and Privacy – Tim Mather
4. Cloud Computing: Principles and Paradigms – Rajkumar Buyya




IAM > Dashboard
Aayush Jha (4592-1296-6899)
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Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
User groups
Users
Roles
Policies
Identity providers
Account settings
Root access management
Temporary delegation requests New
Access reports
Access Analyzer
Resource analysis New
Unused access
Analyzer settings
IAM Dashboard Info
Security recommendations 1
Add MFA for root user
Add MFA for root user - Enable multi-factor authentication (MFA) for the root user to improve security for this account.
Add MFA
Root user has no active access keys
Using access keys attached to an IAM user instead of the root user improves security.
IAM resources
Resources in this AWS Account
User groups 0
Users 0
Roles 3
Policies 0
Identity providers 0
What's new
View all
Updates for features in IAM
AWS STS now supports validation of select identity provider specific claims from Google, GitHub, CircleCI and OCI. 1 month ago
AWS Account
Account ID
459212966899
Account Alias
Create
Sign-in URL for IAM users in this account
https://459212966899.signin.aws.amazon.com/console
Quick Links
My security credentials
Manage your access keys, multi-factor authentication (MFA) and other credentials.
Tools
Policy simulator
The simulator evaluates the policies that you
CloudShell
Feedback
Console Mobile App
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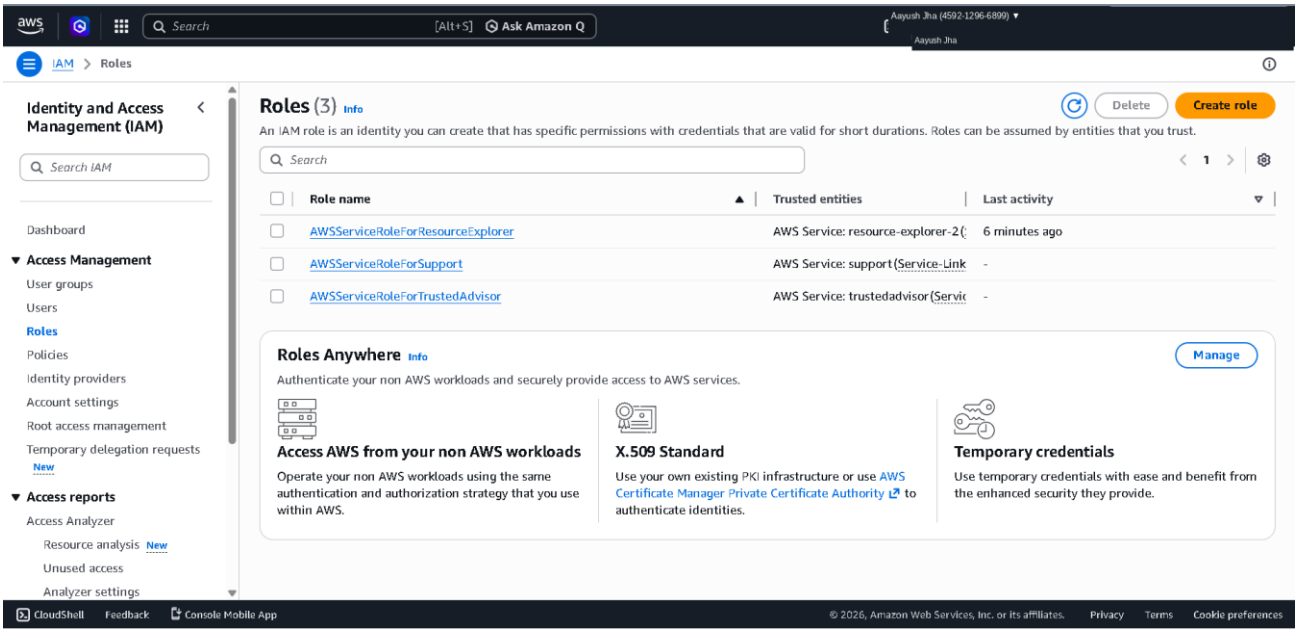
Search
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Ask Amazon Q
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IAM > Roles
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
User groups
Users
Roles
Policies
Identity providers
Account settings
Root access management
Temporary delegation requests New
Access reports
Access Analyzer
Resource analysis New
Unused access
Analyzer settings
Roles (3) Info
Delete
Create role
An IAM role is an identity you can create that has specific permissions with credentials that are valid for short durations. Roles can be assumed by entities that you trust.
Search
Role name
Trusted entities
Last activity
AWSServiceRoleForResourceExplorer
AWS Service: resource-explorer-2
6 minutes ago
AWSServiceRoleForSupport
AWS Service: support (Service-Link
-
AWSServiceRoleForTrustedAdvisor
AWS Service: trustedadvisor (Servic
-
Roles Anywhere Info
Manage
Authenticate your non AWS workloads and securely provide access to AWS services.
Access AWS from your non AWS workloads
Operate your non AWS workloads using the same authentication and authorization strategy that you use within AWS.
X.509 Standard
Use your own existing PKI infrastructure or use AWS Certificate Manager Private Certificate Authority to authenticate identities.
Temporary credentials
Use temporary credentials with ease and benefit from the enhanced security they provide.
CloudShell
Feedback
Console Mobile App
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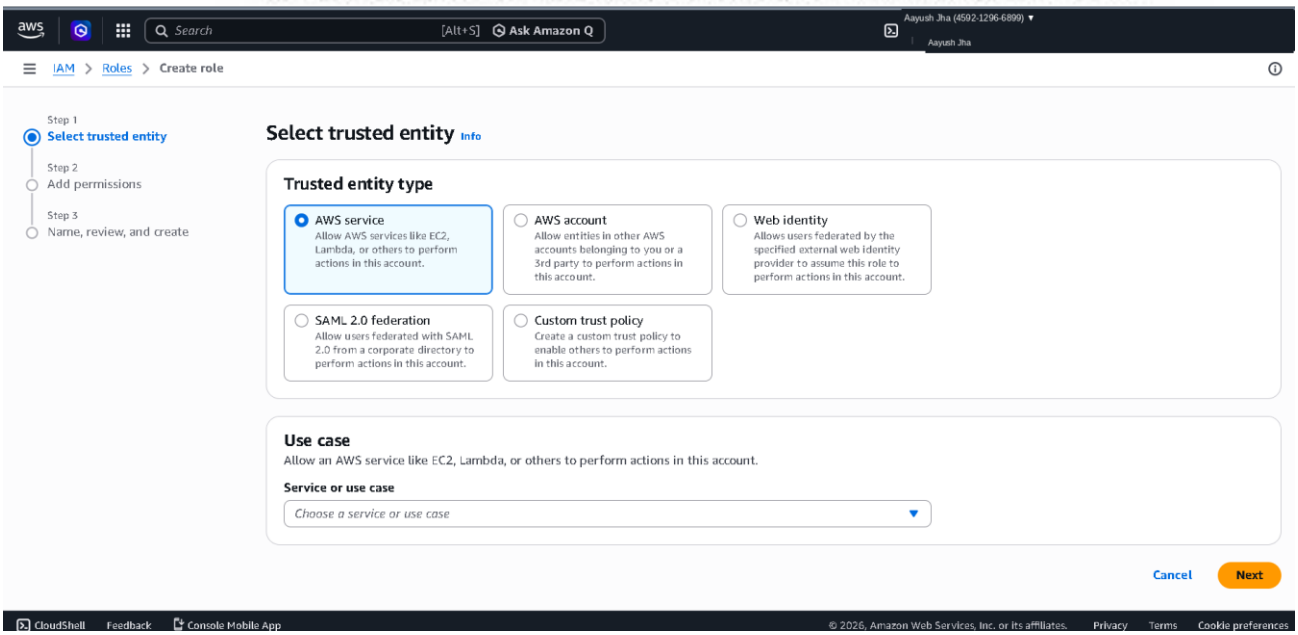
Search [Alt+S] Ask Amazon Q
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IAM > Roles > Create role
Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create
Select trusted entity Info
Trusted entity type
AWS service
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
AWS account
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
Web identity
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
SAML 2.0 federation
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
Custom trust policy
Create a custom trust policy to enable others to perform actions in this account.
Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.
Service or use case
Choose a service or use case
Cancel
Next
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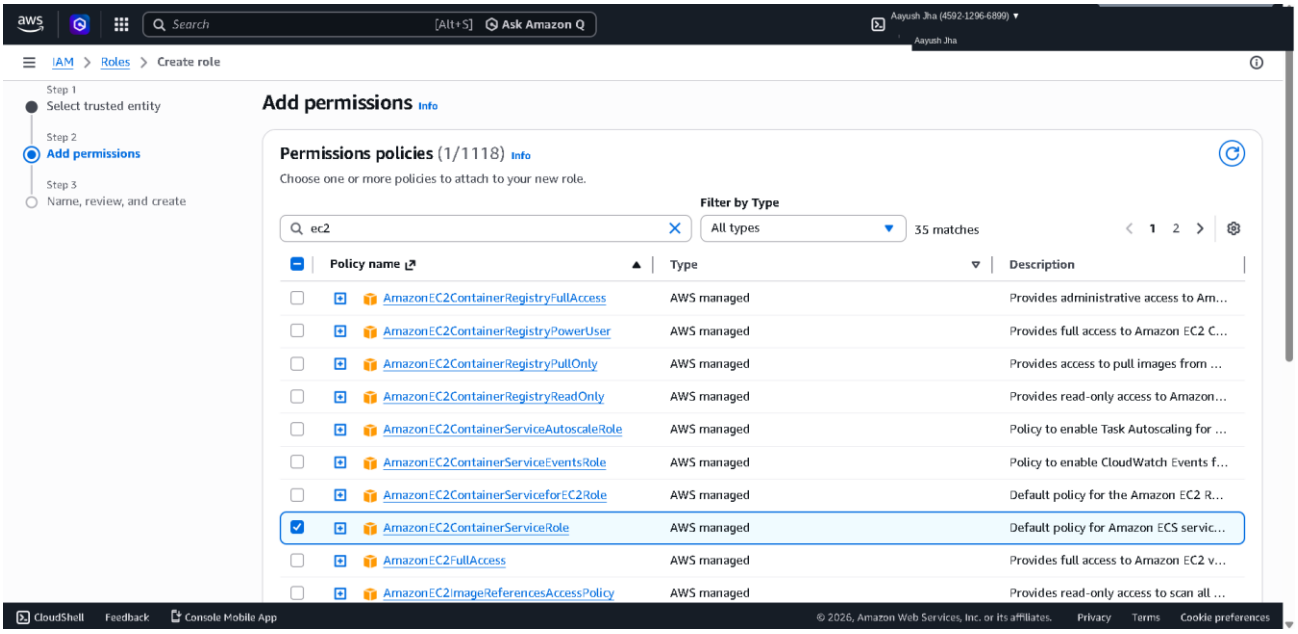
Search [Alt+S] Ask Amazon Q
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IAM > Roles > Create role
Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create
Add permissions Info
Permissions policies (1/1118) Info
Choose one or more policies to attach to your new role.
Filter by Type
ec2
All types
35 matches
1 2
Policy name
Type
Description
AmazonEC2ContainerRegistryFullAccess AWS managed Provides administrative access to Am...
AmazonEC2ContainerRegistryPowerUser AWS managed Provides full access to Amazon EC2 C...
AmazonEC2ContainerRegistryPullOnly AWS managed Provides access to pull images from ...
AmazonEC2ContainerRegistryReadOnly AWS managed Provides read-only access to Amazon...
AmazonEC2ContainerServiceAutoscaleRole AWS managed Policy to enable Task Autoscaling for ...
AmazonEC2ContainerServiceEventsRole AWS managed Policy to enable CloudWatch Events f...
AmazonEC2ContainerServiceforEC2Role AWS managed Default policy for the Amazon EC2 R...
AmazonEC2ContainerServiceRole AWS managed Default policy for Amazon ECS servic...
AmazonEC2FullAccess AWS managed Provides full access to Amazon EC2 v...
AmazonEC2ImageReferencesAccessPolicy AWS managed Provides read-only access to scan all ...
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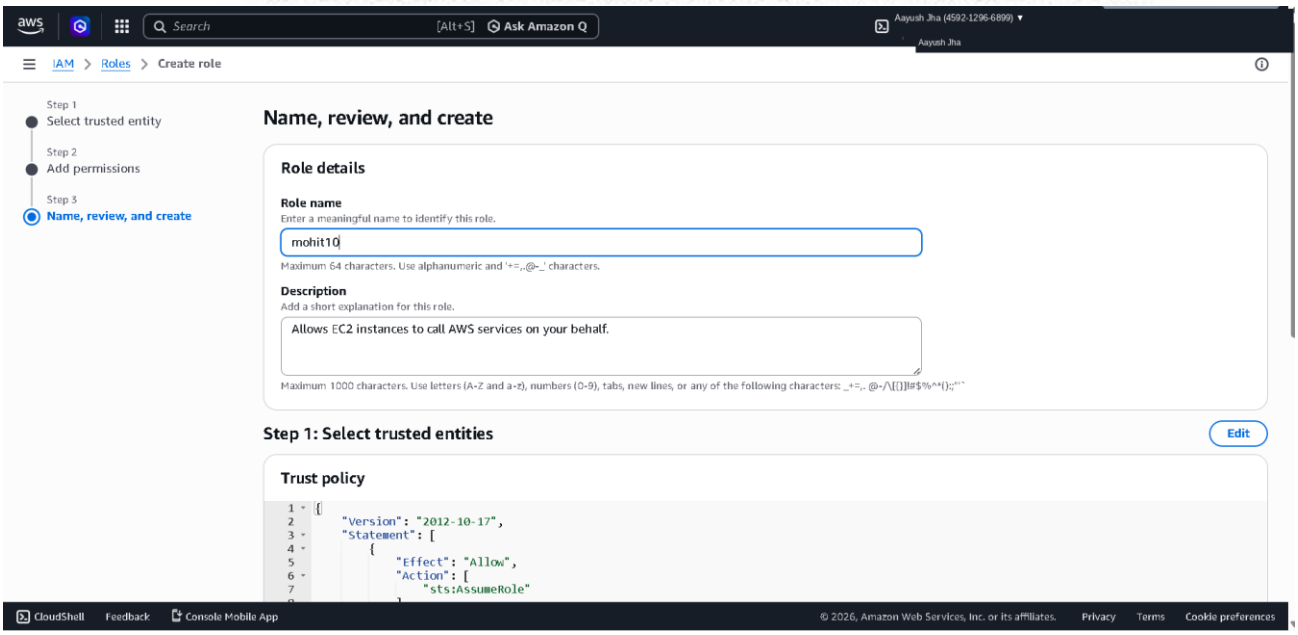
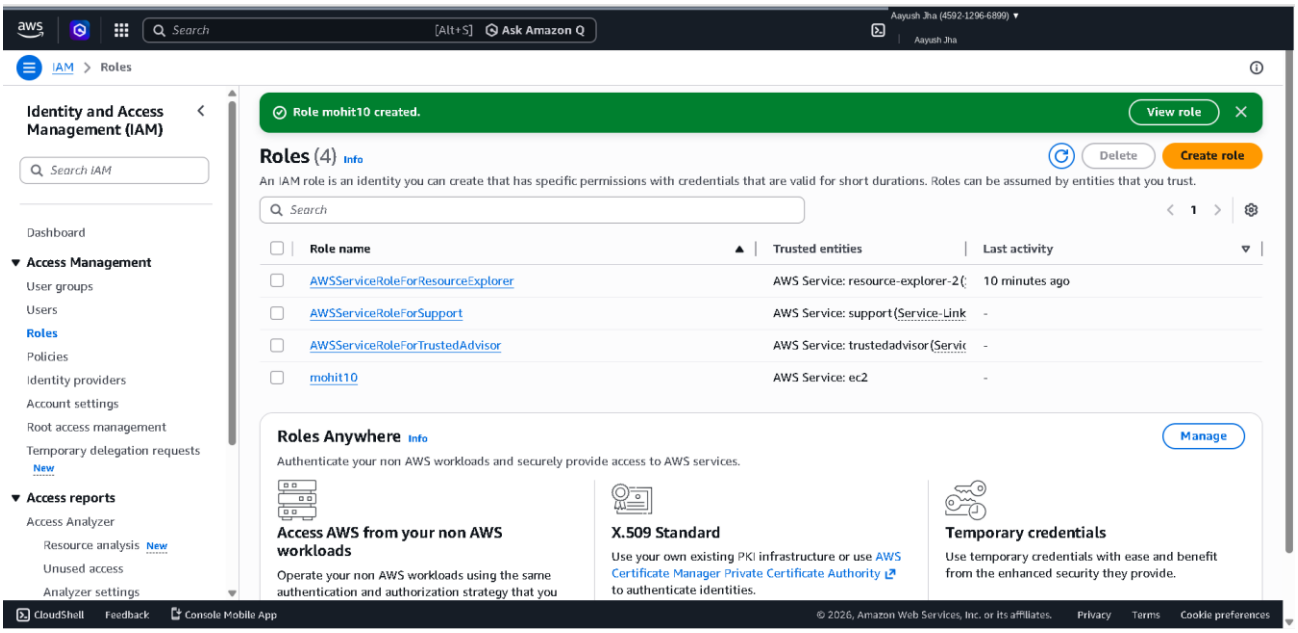

IAM > Roles > Create role
Step 1
Select trusted entity
Step 2
Add permissions
Step 3
Name, review, and create
Name, review, and create
Role details
Role name
Enter a meaningful name to identify this role.
mohit10
Maximum 64 characters. Use alphanumeric and '+=,.@-_' characters.
Description
Add a short explanation for this role.
Allows EC2 instances to call AWS services on your behalf.
Step 1: Select trusted entities
Edit
Trust policy
{
"Version": "2012-10-17",
"Statement": [
{
"Effect": "Allow",
"Action": [
"sts:AssumeRole"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Policies
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Root access management
Temporary delegation requests
New
Access reports
Access Analyzer
Resource analysis New
Unused access
Analyzer settings
Role mohit10 created.
View role
Roles (4) Info
Delete
Create role
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Role name
Trusted entities
Last activity
AWSServiceRoleForResourceExplorer
AWS Service: resource-explorer-2
10 minutes ago
AWSServiceRoleForSupport
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AWSServiceRoleForTrustedAdvisor
AWS Service: trustedadvisor (Servic
mohit10
AWS Service: ec2
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